

ICS 604

Applied Data Science

Department of Information and Computer Sciences
University of Hawai`i at Mānoa

Kyungim Baek

Homework Assignment 3

- Due: **11:59 PM** on **Monday, April 6**
- Please carefully read and follow the assignment instructions and problem requirements.
- Submit only the Jupyter Notebook file (.ipynb) to Lamaku.
 - Do not upload the data file.
 - Make sure to **rename your notebook file** as instructed.

Reminder: Project proposal review

- Two proposals have been assigned.
- Peer feedback is due by **11:59 PM on Monday, March 30.**
 - Write approximately half a page to one page per proposal.
 - Include the title of the project you reviewed.
 - Create a **separate PDF file for each proposal reviewed** and submit it to Lamaku under “Proposal Peer Feedback” in Assignments tool.
 - Name each file using the format:
“**[YourLastName]_feedback_[Proposal_ID].pdf**”
 - Do NOT include your name in the file.
 - Refer to the project requirements document for guidance.
 - Your feedback will be anonymized and shared with the proposal author.

Lecture 17

- Hypothesis Testing: Common Tests
 - χ^2 tests
 - Goodness-of-fit test
 - Test of independence (test of association)

The χ^2 goodness-of-fit test

- A χ^2 test is a hypothesis testing method.
 - One of the oldest hypothesis tests around.
 - The goodness-of-fit χ^2 tests involve checking if **observed frequencies** in one or more categories match expected frequencies.
- Categorical data refers to data with a “nominal scale”.
 - A scale of measurement used to assign events or objects into discrete categories.
 - Similar to an enum in computer science.

The card randomness test

- Ask 200 people to mentally choose 2 cards, one after the other, from an imaginary deck of cards, and look at the suit that people chose.
- Is the choice of the second card random?
 - Do people choose the second independently of the first one?
- Suppose the data for the choice is:

clubs	diamonds	hearts	spades
35	51	64	50

How do we reproduce similar data?

- The observed frequencies $O = (O_1, O_2, O_3, O_4) = (35, 51, 64, 50)$
- How do we sample similar data in Python?
 - Ex: How can you randomly sample 200 observations of 4 events?
- Which distribution should we use?
 - Recall, this is simply a generalization of a coin flip.

```
size = 200
prob = [1/4, 1/4, 1/4, 1/4]
np.random.multinomial(size, prob)

array([45, 47, 55, 53])
```

The χ^2 goodness-of-fit test

- We want to test whether the cards (categories, or whatever other counts) are chosen **uniformly at random**.
 - H_0 : All four suits are chosen with equal probability (each with probability 0.25).
 - H_A : The suits are not all equally likely (i.e., at least one suit has a probability different from 0.25).
- How can we test this hypothesis?

The χ^2 goodness-of-fit test (cont'd.)

- Same as the test we had manually run.
 - Sample uniformly represented cards.
 - Compute some test statistics.
 - Plot where the observed statistics occurs in the distribution under the null hypothesis.
- This is exactly what the χ^2 goodness-of-fit tests.
 - $H_0: P = (0.25, 0.25, 0.25, 0.25)$
 - $H_A: P \neq (0.25, 0.25, 0.25, 0.25)$

The “goodness-of-fit” test statistic

- Similar to what we’ve done via simulation, if we want to test H_0 against H_A , we’re going to need a test statistic.
 - Measures how “close” the data are to the null hypothesis.
 - If the data don’t resemble what you would “expect” to see if the null hypothesis were true, then it probably isn’t true.
 - What are the expected frequencies under the null hypothesis?

```
probabilities = pd.Series([0.25, 0.25, 0.25, 0.25],
                          index=["clubs", "diamonds", "hearts", "spades"])
```

```
expected = 200 * probabilities
expected
```

```
clubs      50.0
diamonds   50.0
hearts     50.0
spades     50.0
dtype: float64
```

Compare expected vs. observed

- Objective: Compare the expected (E_i) with observed values (O_i) in each category.

		♣	♦	♥	♠
expected frequency	E_i	50	50	50	50
observed frequency	O_i	35	51	64	50
difference score	$O_i - E_i$	-15	1	14	0

- Sign should not matter: +14 hearts and -15 clubs should be equivalent to -14 hearts and +15 clubs.
 - Easy: we can just count the squared difference.

Compare expected vs. observed (cont'd.)

- We are also interested in contextualizing that difference.
 - +14 with regards to what?
 - Let's divide by the expectation so we can also represent this difference in terms of units representing how far from the expectation it is.
 - The total distance we are off by is the sum of all of these.

$$\sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

of categories

$$\frac{10}{100} = 0.1$$

$$\frac{10}{1000} = 0.01$$

```
observed = pd.Series([35, 51, 64, 50], index=["clubs", "diamonds", "hearts", "spades"])
observed - expected
```

```
clubs      -15.0
diamonds    1.0
hearts     14.0
spades      0.0
dtype: float64
```

```
(observed - expected)**2 / expected
```

```
clubs      4.50
diamonds    0.02
hearts     3.92
spades      0.00
dtype: float64
```

```
### Goodness of fit statistic
```

```
sum((observed - expected)**2 / expected)
```

8.44 → observed statistic

The χ^2 statistic

- What we computed is the χ^2 statistic: $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$ where k is the number of categories.
- Similar to the simulation-based testing with the tvd distance, it's easy to observe that large value of χ^2 are in favor of the alternatives:
 - If χ^2 is small, then O_i and E_i are close (null hypothesis is okay).
 - If χ^2 is large, then O_i cannot come from the same distribution as E_i .
- Is $\chi^2 = 8.44$ large or small?
- Note that the χ^2 goodness-of-fit test is always a one-sided test.

The sampling distribution of the χ^2 statistic

- What should χ^2 values look like if we sample randomly from the uniform distribution?
 - I.e., what does the sampling distribution of χ^2 looks like under the null hypothesis?
- Recall that, essentially, observed frequency O_i for a category i is distributed according to a binomial $O_i \sim \text{Binomial}(N, P_i)$
- For large values of N and when P_i isn't too close to 0 or 1, the **Central Limit Theorem** says that the binomial is similar to the Gaussian.

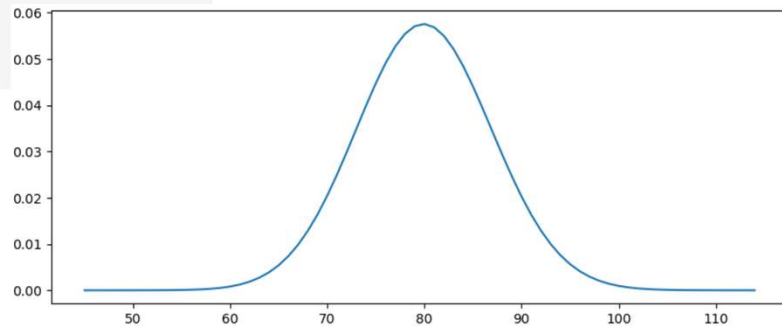

```
N = 200  
p = 0.4
```

```
np.random.binomial(N, p)
```

72

```
x_axis = np.arange(45, 115)  
y_axis = scipy.stats.binom.pmf(x_axis, N, p)
```

```
plt.figure(figsize=(10, 4))  
plt.plot(x_axis, y_axis);
```



From Gaussian to the χ^2

- According to the Central Limit Theorem, when $E_i = N \times P_i$ is large (i.e., N is large and P_i isn't too close to 0 or 1), O_i follows a Gaussian.
- If O_i is normally distributed, then so is $(O_i - E_i)/\sqrt{E_i}$ since E_i is a constant.
 - This represents one of the terms of the statistic (minus the square).
- The test statistic consists of a sum of k terms $(O_i - E_i)^2/E_i$.
- Square and adding a collection of normally distributed variables yields a quantity that has a χ^2 distribution.
 - χ^2 distribution appears in lots of places.
- The parameter of the χ^2 distribution is its degrees of freedom.

Degrees of freedom of the χ^2 distribution

- Square and adding a collection of normally distributed variables yields a quantity that has a χ^2 distribution.
- We are interested in the number of independent “things” we're adding up.
 - Here, only $k - 1$ categories are independent.
 - Since the values sum to a fixed number, then with $k - 1$ categories we can infer the value of the last category.
 - So, the degree of freedom is $k - 1$.

```
scipy.stats.chi2(df=10).rvs()
```

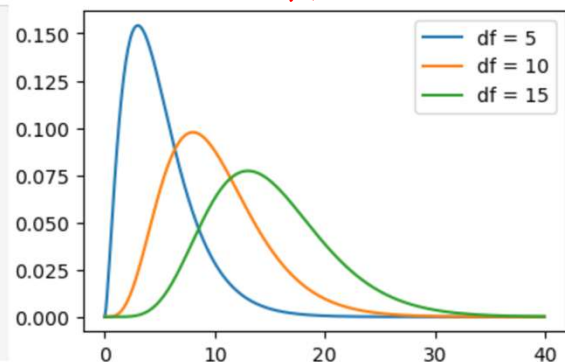
```
4.846586684595898
```

```
scipy.stats.chi2(df=100).rvs(4)
```

```
array([ 99.17736689,  99.9982073, 104.51200853,  85.88606296])
```

```
x_axis = np.arange(0, 40, 0.1)
y_axis = scipy.stats.chi2(df=5).pdf(x_axis)
y2_axis = scipy.stats.chi2(df=10).pdf(x_axis)
y3_axis = scipy.stats.chi2(df=15).pdf(x_axis)

plt.figure(figsize=(4.5, 3))
plt.plot(x_axis, y_axis, label="df = 5")
plt.plot(x_axis, y2_axis, label="df = 10")
plt.plot(x_axis, y3_axis, label="df = 15")
plt.legend();
```

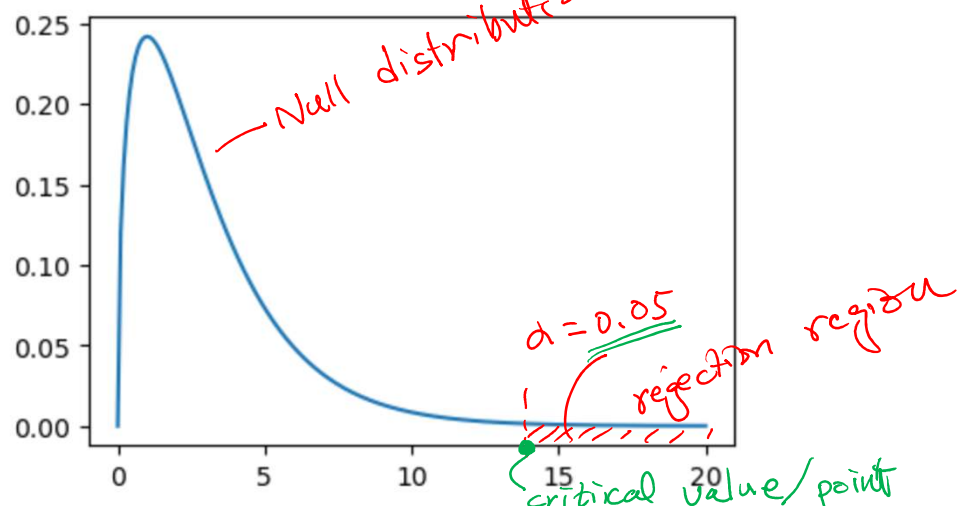


Finding the rejection region

- What values of the test statistic (i.e., χ^2 statistic) lead to rejecting the null hypothesis?
- Recall that
 - if χ^2 is small, then O_i and E_i are close (null hypothesis is okay).
 - if χ^2 is large, then O_i cannot come from the same distribution as E_i .

```
x_axis = np.arange(0, 20.1, 0.1)
y_axis = scipy.stats.chi2(df=3).pdf(x_axis)

plt.figure(figsize=(4.5, 3))
plt.plot(x_axis, y_axis);
```



```
chi2_rv = scipy.stats.chi2(df=3)
```

```
### The cdf refers to the cumulative distribution function.
### The cumulative distribution function (cdf) is the probability
### that the variable takes a value less than or equal to x.
```

```
chi2_rv.cdf(0)
```

```
0.0
```

```
chi2_rv.cdf(20)
```

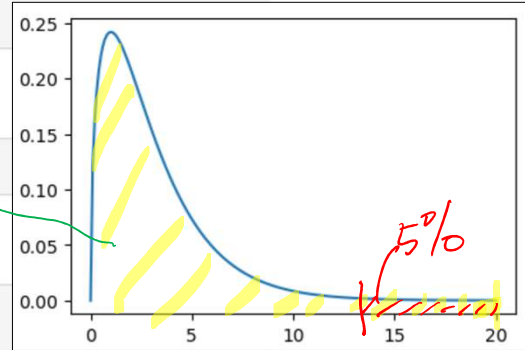
```
0.9998302575644472
```

```
print(x_axis[:10])
```

```
print(x_axis[-10:])
```

```
[0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9]
```

```
[19.1 19.2 19.3 19.4 19.5 19.6 19.7 19.8 19.9 20. ]
```



```
probs = chi2_rv.cdf(x_axis)
```

```
print(np.round(probs[:10], decimals=4))
```

```
print(np.round(probs[-10:], decimals=4))
```

```
[0.  0.0082 0.0224 0.04  0.0598 0.0811 0.1036 0.1268 0.1505 0.1746]
```

```
[0.9997 0.9998 0.9998 0.9998 0.9998 0.9998 0.9998 0.9998 0.9998 0.9998]
```

```
np.where(probs > 0.95)
```

```
(array([ 79,  80,  81,  82,  83,  84,  85,  86,  87,  88,  89,  90,  91,
        92,  93,  94,  95,  96,  97,  98,  99, 100, 101, 102, 103, 104,
       105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117,
       118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130,
       131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143,
       144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156,
       157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169,
       170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182,
       183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195,
       196, 197, 198, 199, 200], dtype=int64),)
```

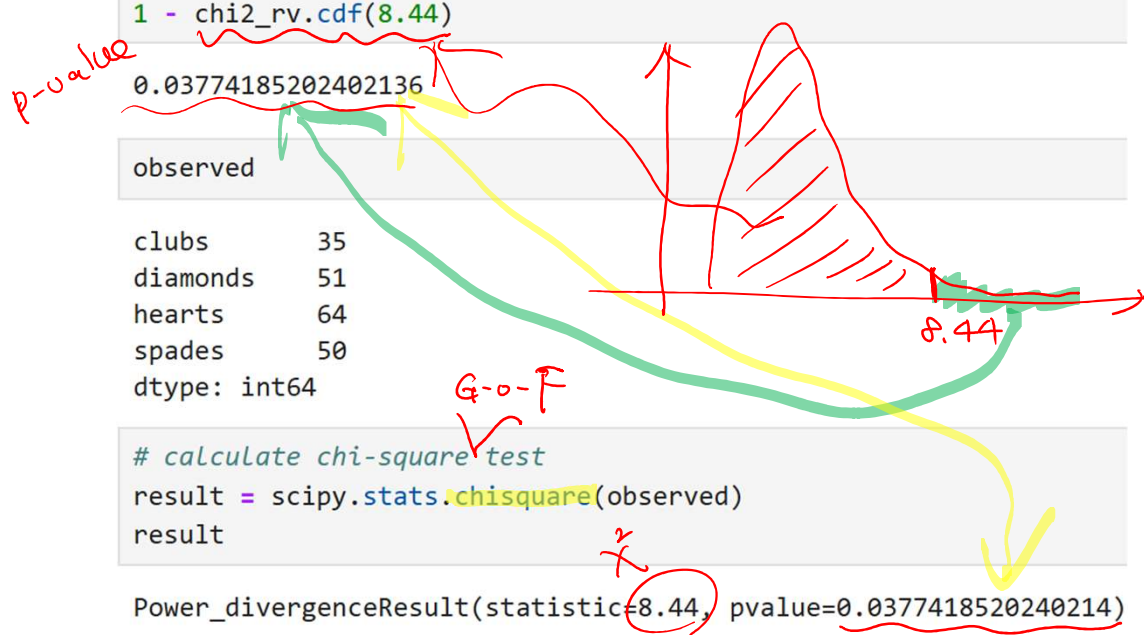
79

7.9

critical value

7.814727903251179

```
Power_divergenceResult(statistic=8.44, pvalue=0.0377418520240214)
```



observed

```
clubs      35
diamonds   51
hearts     64
spades     50
dtype: int64
```

```
### Specifying different null hypothesis
```

```
result = scipy.stats.chisquare(f_obs=observed, f_exp=[40, 50, 60, 50])
result
```

```
Power_divergenceResult(statistic=0.9116666666666666, pvalue=0.8226115000538757)
```

```
print("chi-squared statistic =", result[0])
print("p-value =", round(result[1], 4))
```

```
chi-squared statistic = 0.9116666666666666
p-value = 0.8226
```

χ^2 observed

result.statistic
result.pvalue

The χ^2 test of independence (or association)

- Extends the goodness-of-fit framework to situations involving two categorical variables.
 - Determine whether two categorical variables are independent of each other (or if there is a relationship between them).
- Example:
 - Support you would like to know whether a site is more conducive to growth of some coral species.
 - You conduct a test where you introduce species to site A and site B, and observe whether they live, die, or become unhealthy.

Example (cont'd.)

- 180 entries introduced in site A or site B.
- 87 in site A and 93 in site B.

contingency table

var 2

Condition	siteA	siteB	Total
dies	30	13	43
unhealthy	44	65	109
lives	13	15	28
Total	87	93	180

var 1

- **Question:** Is the distribution of the data the same across both sites?
- Are the proportions of lives, dies, or unhealthy the same across both sites? (Or is one site more conducive to coral health?)

Constructing the hypothesis test

- O_{ij} is a count (observed frequency) of the number of respondents that are at a site (site A for $j = 1$ or site B for $j = 2$), which lead to condition i (lives, dies, unhealthy).
- The total number of observations is written as N .
- R_i to denote the row totals.
- C_j to denote the column totals.

$$(2-1)(3-1) = 2$$

$\hat{p}_i = \frac{R_i}{N}$

proportion under null hyp.

	SiteA	SiteB	
dies	O_{11} E_{11}	O_{12} E_{12}	R_1
Unhealthy	O_{21} $\vdots$	O_{22} $\vdots$	R_2
lives	O_{31} E_{31}	O_{32} E_{32}	R_3
	C_1	C_2	N

$\hat{p}_1$
 $\hat{p}_2$
 $\hat{p}_3$

Constructing the hypothesis test (cont'd.)

- If both sites yield the same outcomes, then probability of dies is equal in both sites.
 - By the same token, probability of unhealthy is also similar in both sites.
 - By the same token, probability of lives also similar in both sites.
- H_0 : All of the following are true:
 - $P_{11} = P_{12}$ (same probability of dies)
 - $P_{21} = P_{22}$ (same probability of unhealthy)
 - $P_{31} = P_{32}$ (same probability of lives)

Constructing the hypothesis test (cont'd.)

- We denote P_i , where i in $\{1, 2, 3\}$ as the probability of dies, unhealthy or lives, respectively.
- Unlike goodness of-fit-test, the null hypothesis doesn't actually specify a particular value for P_i , but we can estimate it from the data.

$$\hat{P}_i = \frac{R_i}{N}$$

- E.g., if 43 out of 180 species died, then the empirical probability is simply 43/180 (= 0.24).
- Expected frequency assuming independence of site and outcome is therefore: $\hat{E}_{ij} = \frac{R_i}{N} \times C_j$

Defining the test statistic

- Following the exact same strategy that we used in the goodness-of-fit test:

- For a contingency table with r rows and c columns, the equation that defines our χ^2 statistic is:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(\hat{E}_{ij} - O_{ij})^2}{\hat{E}_{ij}}$$

- For the same reasons as earlier, this statistic is also distributed according to the χ^2 distribution.
 - How many independent points are we analyzing?
 - Here, $df = (r - 1)(c - 1)$.

Assumptions of the test

- Expected frequencies are sufficiently large.
 - Expected frequencies larger than at least 5.
 - For larger tables, at least 80% of the expected frequencies are above 5 and none is 0.
 - Assumption needed to approximate the Central Limit Theorem.
- Data are independent of one another.
 - E.g., the fact that a species dies, does not increase the change of another species dying.

Chi-squared test exercises

- Complete the exercise:
`exercise4_chi_squared_tests.ipynb`
- Submit your file to “Exercises” under Assignments in Lamaku.
- Due: **Monday, March 30**
- Not graded but counts toward class participation.